

Mu (Henry) Ha

Vancouver, BC | m8ha@uwaterloo.ca | 519-503-5609 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Manufacturing Data Analyst | Predictive Maintenance | Production Analytics

Eligible for TN visa sponsorship under USMCA (Engineer / Computer Systems Analyst categories)

Manufacturing: SPC (Cp/Cpk), Yield Analysis (FPY, FTY), Root Cause Analysis, Test Plan Development, GD&T, Tolerance Analysis, SAP BOM/Routings, Cost Analysis, Quality Control, Lean Manufacturing

Data Analytics: Python, C/C++, SQL, Pandas, Time Series Data, Dashboarding, Data Validation, AWS, Git/GitHub

System Tools: Sensor Integration, Simulink, PLCs, TCP/IP, PID, Sensor Fusion, MATLAB, AutoCAD, CI/CD, Docker

EXPERIENCE

Brilliant Automation (Predictive Maintenance Capstone)

Apr 2025 – Jun 2025

Systems Data Engineer

Vancouver, BC

- Built a predictive maintenance and monitoring system processing multi-rate telemetry and waveform inputs to generate interpretable health scores across 12 machine metrics for production monitoring.
- Implemented leakage-free expanding-window validation (5 splits, N=3,045) with structured out-of-fold evaluation to assess performance under deployment-like conditions.
- Developed monitoring dashboards to track machine performance, anomaly trends, and system-level metrics.

Weir Minerals (Internship)

Sep 2019 – Apr 2020

Manufacturing Engineer

Surrey, BC

- Estimated production costs and created BOMs & routings in SAP to support production planning and control.
- Conducted mechanical validation testing (gasket compression, deformation) and led root cause investigations to resolve quality deviations and ensure spec compliance.
- Improved cost estimation workflow by ~30% through data standardization, enhancing traceability & planning.

Weir Minerals (Internship)

Jan 2019 – Apr 2019

Design Engineer

Toronto, ON

- Designed mechanical assemblies with GD&T to ensure manufacturability and tolerance robustness.
- Collaborated with fabrication teams to resolve tolerance stack-ups and support production validation.
- Performed design verification including material selection, seal geometry checks, and stress validation.

Resolution Life US (Internship)

Sep 2021 – Apr 2022

AWS Cloud Engineer

Toronto, ON

- Built standardized EC2 golden images to support consistent environments, reducing deployment time by 80%.
- Automated AWS usage dashboards for cost and trend analysis, supporting data-driven infrastructure decisions.

University of Waterloo (Internship)

Jan 2021 – Apr 2021

Data Scientist Research Assistant

Toronto, ON

- Implemented and trained U-Net neural network architectures for semantic segmentation of satellite imagery.
- Achieved 0.026 validation MSE and 90% accuracy by benchmarking models and hyperparameter tuning.

PROJECT

Manufacturing Test Automation Platform

Jul 2025 – Jan 2026

- End-to-end manufacturing test and yield analytics platform simulating production environments.
- Developed a TCP/IP-based DUT simulator to emulate manufacturing test communication, control, and faults.
- Designed structured acceptance and qualification test plans with defined coverage mapping and retry logic.
- Implemented an automated test framework supporting multi-SN execution, structured JSON logging, fault injection (timeouts, drift, intermittent failure), and traceable result storage.
- Built yield analytics modules (FPY, FTY, Pareto, stratification) and deployed a Dockerized, CI-enabled validation pipeline with auto-generated qualification reports.

EDUCATION

University of British Columbia

Sep 2024 – Jul 2025

Master of Data Science

University of Waterloo

Sep 2018 – Apr 2024

B.Sc. Mechanical Engineering (*Honours, Co-op*) AI Option, Management Science Option